

NORFOLK, NE, USA

WMO: 725560

Lat: 41.986N

Lon: 97.435W

Elev: 473

StdP: 95.77

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14941

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.9	-18.1	-25.1	0.4	-20.3	-22.5	0.5	-17.5	15.2	-7.7	13.8	-7.2	4.3	320	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	34.3	23.5	32.6	23.0	31.1	22.3	25.4	31.3	24.6	30.4	23.6	29.2	6.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	19.7	29.2	22.7	18.5	28.3	21.8	17.4	27.4	81.0	31.5	76.8	30.3	73.0	29.0	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.5	11.1	9.9	-25.4	36.8	2.9	2.1	-27.5	38.3	-29.1	39.5	-30.8	40.7	-32.8	42.2		
			-25.7	27.0	2.8	0.7	-27.7	27.5	-29.3	27.9	-30.9	28.3	-32.9	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	-5.0	-3.0	3.6	9.7	15.9	21.5	24.1	22.7	18.5	10.9	3.3	-3.5
	DBStd	11.66	7.09	7.00	6.83	5.62	4.87	3.91	3.34	3.18	4.76	5.39	5.91	6.40
	HDD10.0	1816	467	366	219	71	8	0	0	0	2	53	212	418
	HDD18.3	3594	725	598	458	263	106	16	2	4	56	236	452	677
	CDD10.0	1800	0	1	19	62	192	346	438	393	256	82	11	0
	CDD18.3	536	0	0	1	4	32	112	182	139	60	7	0	0
	CDH23.3	5571	0	0	17	85	385	1154	1921	1281	634	90	3	0
	CDH26.7	2154	0	0	3	25	129	429	833	484	223	27	0	0
Wind	WSAvg	4.6	4.8	5.0	5.2	5.5	4.9	4.4	3.9	3.7	4.3	4.7	4.7	4.6
Precipitation	PrecAvg	668	15	20	41	65	98	116	79	77	62	48	28	20
	PrecMax	961	44	81	185	190	219	310	229	204	206	173	101	74
	PrecMin	371	2	1	1	6	10	14	0	14	9	1	0	1
	PrecStd	129	11	15	34	38	50	68	48	45	41	38	26	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.4	18.4	25.6	30.1	33.2	34.9	37.4	36.0	33.8	30.2	22.9	14.2
		MCWB	7.1	9.5	13.4	16.3	19.4	23.2	23.8	23.7	21.9	18.5	12.8	7.7
	2%	DB	10.1	12.9	21.5	25.7	30.0	32.7	34.8	33.2	31.5	25.5	19.0	10.8
		MCWB	4.9	7.3	12.4	15.0	19.2	22.7	24.0	24.0	21.2	16.2	11.3	6.0
	5%	DB	6.8	9.8	17.6	22.3	27.7	31.0	32.9	31.1	29.3	22.9	15.5	7.8
		MCWB	3.0	5.3	11.1	13.5	18.0	21.9	24.0	23.1	20.4	14.8	9.4	3.8
	10%	DB	4.2	6.8	14.3	19.6	25.0	29.2	31.2	29.4	27.1	20.2	12.4	5.0
		MCWB	1.6	3.5	8.8	11.9	16.8	20.9	23.3	22.2	19.2	13.5	7.6	2.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	10.8	16.4	18.3	22.2	25.3	27.1	26.1	23.9	20.2	15.0	9.4
		MCDB	14.2	15.8	21.4	26.9	28.5	31.5	33.0	32.1	30.0	26.5	19.2	12.8
	2%	WB	5.1	7.8	14.0	16.2	20.6	24.1	25.9	25.0	22.7	18.0	12.0	5.9
		MCDB	9.7	12.8	19.2	23.5	27.6	30.4	31.7	30.9	28.8	23.0	17.4	10.1
	5%	WB	3.3	5.4	11.4	14.4	19.1	23.1	25.0	24.0	21.6	16.1	9.9	4.1
		MCDB	6.4	9.5	17.4	21.2	25.6	29.2	30.8	29.5	27.4	21.0	15.2	7.2
	10%	WB	1.7	3.5	8.9	12.7	17.7	22.0	24.0	23.1	20.4	14.3	7.7	2.4
		MCDB	4.0	6.6	14.2	18.7	23.8	27.6	29.6	28.2	25.8	19.1	12.5	5.0
Mean Daily Temperature Range	5% DB	MDBR	10.6	10.7	12.2	12.5	12.3	12.0	11.7	11.6	13.2	12.9	11.8	9.9
		MCDBR	14.9	15.3	18.3	17.6	15.9	13.8	13.3	13.4	14.9	17.0	17.3	14.2
		MCWBR	10.2	10.1	10.4	9.0	7.3	6.2	5.9	6.2	6.8	8.7	10.1	9.8
	5% WB	MCDBR	14.2	14.6	16.4	15.7	13.5	12.0	11.4	11.9	12.7	13.6	15.7	13.3
		MCWBR	9.9	9.9	9.7	8.7	7.0	6.4	6.1	6.2	6.5	7.9	9.9	9.6
Clear-Sky Solar Irradiance	taub	0.275	0.288	0.313	0.378	0.397	0.411	0.414	0.415	0.370	0.323	0.285	0.274	
	taud	2.453	2.455	2.426	2.253	2.278	2.299	2.298	2.285	2.372	2.488	2.509	2.465	
	Ebn at Noon	884	926	937	890	878	864	857	844	862	872	866	853	
	Edh at Noon	75	88	102	131	131	129	128	126	107	84	70	68	
All-Sky Solar Radiation	RadAvg	2.04	2.97	3.91	4.99	5.83	6.59	6.64	5.64	4.57	3.21	2.25	1.67	
	RadStd	0.17	0.21	0.32	0.32	0.41	0.46	0.34	0.41	0.31	0.36	0.16	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page